

$$\text{rel freq} = \text{freq}/n, \text{ density} = (\text{rel freq})/w$$

$$k = \sqrt{n} \quad w = (\text{max} - \text{min})/k$$

num and width of freq table classes

$$P_k = x_{\frac{kn}{100}} \quad \text{kn/100th term in sorted data}$$

$$\text{IQR} = Q_3 - Q_1 \quad \text{each qrtl has 25\% of data}$$

$$\text{Var} : s^2 = \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - n\bar{x} \right)$$

$$\text{SD} : s = \sqrt{s}$$

$$P(E) = \# E \text{ occs.} \div \# S \text{ occs.}$$

given all outcomes in S are equally likely

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\text{Disjoint} : P(A \cap B) = 0 \quad \text{mutually exclusive}$$

$$\text{Indpndnt} : P(A \cap B) = P(A)P(B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A|B)P(B) \vee P(B|A)P(A)$$

$$P(A) = P(A|B)P(B) + P(A|\bar{B})P(\bar{B})$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A|B)P(B)}{P(A)}$$

$$E(X) = \sum_{i \in R(X)} iP(X = i) = \mu$$

$$\text{Var}(X) = E(X^2) - \mu^2 = \sigma^2$$

$$E(X)^2 = \sum_{i \in R(X)} i^2 P(X = i)$$

$$\text{B:} \begin{cases} R(x) &= \{0, 1, \dots, n\} \\ P(X = j) &= \binom{n}{j} p^j (1-p)^{n-j} \\ &: \text{Prob of } j \text{ succ. out of } n \text{ trials} \\ E(X) &= np = \mu \\ \text{Var}(X) &= \sqrt{np(1-p)} = \sigma \end{cases}$$

$$\text{Poi:} \begin{cases} R(x) &= \{0, 1, 2, 3, 4, \dots\} \\ P(X = j) &= \frac{e^{-\lambda} \lambda^j}{j!} \\ &: \text{Prob of } j \text{ occ. in interv.} \\ E(X) &= \lambda \\ \text{Var}(X) &= \lambda \quad (SD = \sqrt{\lambda}) \end{cases}$$

$$\text{G:} \begin{cases} R(x) &= \{1, 2, 3, 4, \dots\} \\ P(X = j) &= (1-p)^{j-1} p, \quad 0 < p < 1 \\ E(X) &= 1/p \\ \text{Var}(X) &= (1-p)/p^2 \end{cases}$$

HG: $\begin{cases} R(x) &= \max(0, n+m-N), \dots, \min(m, n) \\ P(X=j) &= \frac{\binom{m}{j} \binom{N-m}{n-j}}{\binom{N}{n}} \\ E(X) &= \frac{nm}{N} \\ Var(X) &= \frac{nm(N-n)(N-m)}{N^2(N-1)} \end{cases}$	Emp. Rule: $\bar{x} \pm k \times \text{SD}$ (64:1, 95:2, 99.7:3) Zsco: $z = \frac{x - \bar{x}}{s}$ (>3isout) $\mu_{\bar{X}} = \mu$ $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$ $\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$ (givOrCLT)
Co: $\begin{cases} f(x) &\geq 0 \forall x \in R(X) \\ \int_{R(X)}^a f(x)dx &= 1 \\ P\{a < X < b\} &= \int_a^b f(x)dx \\ E(X) &= \int_{R(X)} xf(x)dx = \mu \\ Var(X) &= E(X^2) - \mu^2 \\ f(X &= a) \neq prob! \end{cases}$	B: trials independent and have exactly 2 outcomes, prob of success same $\forall$ trials, where n fixed in advance Poi: for prob of # occs in interval (X); occs: indpdnt, not fixed, expctd #/interval constant (λ), total # not fixed
Exp: $\begin{cases} f(x) &= \lambda e^{-\lambda x} \text{ where } x \geq 0 \\ E(x) &= \frac{1}{\lambda} \\ Var(x) &= \frac{1}{\lambda^2} \end{cases}$	G: # of trials till 1st success with prob. of success p ; trials indpdnt HG: X : # w/ trait in n draws w/o replcmnt; m of N have trait
Norm: $\begin{cases} f(x) &= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} \\ \text{Mean} &= \mu \text{ (location), SD} = \sigma \text{ (shape)} \end{cases}$ $Z_{\alpha} = 100(1 - \alpha)^{\text{th}} \text{prcnt}$ $(X - \mu) \div \sigma = Z \sim N(0, 1)$ $\sigma Z + \mu = X \sim N(\mu, \sigma)$	Exp: time till event; λ : mean # occs/unit intrvl E(X): mean of r.v.; cont. analogue is pdf BvP: B: finite trials, poi: inf possible occs tip: On cont, esp poi, remember complementing to get finite addition problems

$$\bar{X} = \hat{\mu}, s = \hat{\sigma}, SE_{\bar{X}} = s \div \sqrt{n}$$

$$n = (z_{\alpha/2} \sigma \div SE)^2$$

$$\text{C.I. of } \mu: \bar{X} \pm t_{\alpha/2, n-1} SE_{\bar{X}}$$

1 Sample

1 set μ, H, α ; 2 check CLT / norm; 3 calc $\bar{X}, s$; then T

$$T = \frac{\bar{X} - c}{s/\sqrt{n}} \sim t(n-1)$$

$$\text{Reject } H_0 \text{ if: } \begin{cases} H_1 : u \neq c & |T| > t_{\alpha/2, n-1} \\ & p = 2P(t > |T|) \\ H_1 : u > c & T > t_{\alpha, n-1} \\ & p = P(t > T) \\ H_1 : u < c & T < -t_{\alpha, n-1} \\ & p = P(t < T) \end{cases}$$

2 Indpndnt Samples

$$\sigma_1 = \sigma_2 \text{ if } \frac{\max(s_1, s_2)}{\min(s_1, s_2)} < 2$$

$$\sigma_1 = \sigma_2 : \begin{cases} T = \frac{\bar{X}_1 - \bar{X}_2}{SE_{\text{pooled}}} \\ df^* = n_1 + n_2 - 2 \end{cases}$$

$$\sigma_1 \neq \sigma_2 : \begin{cases} T = \frac{\bar{X}_1 - \bar{X}_2}{SE_{\text{unpooled}}} \\ df^* = \min(n_1 - 1, n_2 - 1) \end{cases}$$

$$SE_{u..} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, SE_{p..} = \sqrt{S_{p..}^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$S_{p..}^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$\text{Reject } H_0 \text{ if: } \begin{cases} H_1 : \mu_1 - \mu_2 \neq 0 & |T| > t_{\alpha/2, df} \\ H_1 : \mu_1 - \mu_2 > 0 & |T| > t_{\alpha, df} \\ H_1 : \mu_1 - \mu_2 < 0 & |T| < -t_{\alpha, df} \end{cases}$$

$$\text{indp C.I.: } (\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2, df} SE?$$

accept H_0 if has zero

2 paired

$$\text{Paired: } \begin{cases} H_0 : \text{use } \mu_d \\ d_i = x_{1i} - x_{2i} \forall i = 1, \dots, n \\ \text{compute } \bar{d}, s_d, SE_{\bar{d}} \\ SE_{\bar{d}} = s_d / \sqrt{n} \\ T = \frac{\bar{d}}{SE_{\bar{d}}} \sim t(n-1) \end{cases}$$

$$\text{Reject } H_0 \text{ if: } \begin{cases} H_1 : \mu_d \neq 0 & |T| > t_{\alpha/2, n-1} \\ H_1 : \mu_d > 0 & T > t_{\alpha, n-1} \\ H_1 : \mu_d < 0 & T < -t_{\alpha, n-1} \end{cases}$$

$$\text{pair C.I.: } \bar{d} \pm t_{\alpha/2, n-1} SE_{\bar{d}}$$

accept H_0 if has zero

pop param

$$\hat{p}: \begin{cases} \hat{p} &= \frac{X}{n} \\ \hat{p} &\sim N(p, \sqrt{p(1-p)/n}) \end{cases}$$

X : # of outcmts w/ a char.

1 set p, H, α ; 2 check np and $n(1-p) \geq 5$; 3 calc $\hat{p}$; then Z

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \sim N(0, 1)$$

$$p = p_0 \text{ under } H_0 \text{ Reject } H_0 \text{ if: } \begin{cases} H_1 : p \neq p_0 & |Z| > z_{\alpha/2} \\ & 2P(z > |Z|) < \alpha \\ H_1 : p > p_0 & Z > Z_{\alpha} \\ & P(z > Z) < \alpha \\ H_1 : p < p_0 & Z < -Z_{\alpha} \\ & P(z < Z) < \alpha \end{cases}$$

$$\text{C.I. for } p: \hat{p} \pm Z_{\alpha/2} \sqrt{\hat{p}(1-\hat{p})/n}$$

sign test for median

(1) set H, α , (2) get $X_i - M_0$'s and their signs (rm=0)

$$\text{Reject } H_1 \text{ if } p < \alpha: \begin{cases} H_1 : M \neq M_0 & K = N_s \\ & 2P(K \leq K_{\text{obs}}) \\ H_1 : M > M_0 & K = N_+ \\ & P(K \geq K_{\text{obs}}) \\ H_1 : M < M_0 & K = N_- \\ & P(K \geq K_{\text{obs}}) \end{cases}$$

N_+ : # of pos, N_- : # of neg, $N_s = \min(N_+, N_-)$

$$K \sim B(n, 0.5) = \frac{n!}{k!(n-k)!} (0.5^n)$$

sign test for paired

get sign of d_i 's instead, rm $d_i = 0$

$$\begin{cases} H_1 : \mu_d \neq 0 & K = N_s \\ & 2P(K \leq K_{\text{obs}}) \\ H_1 : \mu_d > 0 & K = N_+ \\ & P(K \geq K_{\text{obs}}) \\ H_1 : \mu_d < 0 & K = N_- \\ & P(K \geq K_{\text{obs}}) \end{cases}$$

$|\rho| \rightarrow 1$: more linear (range $[-1, 1]$)

$$r = S_{XY} / (\sqrt{S_{XX}} \sqrt{S_{YY}})$$

$$S_{XX} = \sum x_i^2 - n\bar{x}^2$$

$$S_{YY} = \sum y_i^2 - n\bar{y}^2$$

$$S_{XY} = \sum x_i y_i - n\bar{x}\bar{y}$$

$$\hat{y}_i = b_0 + b_1 x_i = \begin{cases} b_0 &= S_{XY} / S_{XX} \\ b_1 &= \bar{y} - b_1 \bar{x} \end{cases}$$

predict only in range of params